



Autonomous characterisation of fouling-free feasible operating regions in continuous flow precipitation of ketoprofen nanoparticles

Ishaa Mane¹, Arun Pankajakshan¹, Sayan Pal¹, Maximilian O. Besenhard¹, Asterios Gavriilidis¹, Federico Galvanin^{1,*}, Luca Mazzei^{1,*}

University College London, Department of Chemical Engineering, London, WC1E 7JE, United Kingdom

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ABSTRACT

Continuous antisolvent precipitation enables efficient pharmaceutical manufacturing but is susceptible to fouling, which disrupts operation and compromises product quality. To address this challenge, we developed and validated a closed-loop, autonomous screening platform for identifying non-fouling conditions in continuous antisolvent precipitation, demonstrated for ketoprofen nanoparticle formation in a confined impinging jet reactor using ethanol as solvent, water as antisolvent, and Kollidon VA 64 (PVP/VA) as additive. The platform integrates real-time, pressure drop-based fouling detection with adaptive experimental design. Initially, it performs design of experiments to generate feasibility labels (fouling or non-fouling). A class balancing strategy then targets under-represented regions, and active learning with Gaussian Process (GP) classification selects subsequent experiments to reduce uncertainty in feasibility prediction. With an experimental budget of 50 precipitation experiments, the system autonomously mapped the boundary separating fouling and non-fouling regions across three key process variables: total volume flow rate (1–11 mL min⁻¹), antisolvent-to-solvent volume flow rate ratio (4–10), and drug-to-additive mass ratio (0.2125–4.25). The resulting GP classifier achieved high prediction accuracy with an area under the receiver operating characteristic curve (AUC) greater than 0.8 in cross-validation experiments and an AUC of 1.0 in a set of eight follow-up validation experiments. The methodology revealed complex interactions between process conditions and fouling behaviour, identifying robust non-fouling operating windows. This autonomous framework provides a systematic and efficient approach for fouling detection and operating region characterisation in continuous pharmaceutical precipitation, reducing development effort and improving process reliability.

1. Introduction

Continuous antisolvent precipitation has gained significant attention in pharmaceutical manufacturing for its ability to produce Active Pharmaceutical Ingredient (API) nanoparticles that offer improved bioavailability and therapeutic performance [1–4]. Continuous systems, such as confined impinging jet reactors (CIJR) [5], mixed-suspension mixed-product removal (MSMPR) crystallisers [6], tubular reactors [6], and oscillatory flow reactors [3], offer steady-state operation, improving process control and reducing solvent consumption.

Among these, CIJR stand out for their ability to induce rapid micromixing and generate high local supersaturation, which is critical for homogeneous conditions, initiating fast nucleation and forming nanoparticles in antisolvent systems [7–9]. These rapid precipitation dynamics occur within confined volumes and over short timeframes, potentially promoting fouling due to surface crystallisation, particle accumulation, and particle agglomeration. The first leads to reactor

wall deposits, the second clogs flow channels, while the third promotes uncontrolled particle growth [10–12]. As these processes occur, they disrupt operations, complicate cleaning, and reduce efficiency. In addition to operational issues, these fouling mechanisms affect nanoparticle attributes, compromising size distribution, morphology, and stability.

In confined continuous-flow crystallisers and microreactors, clogging and encrustation have been widely reported. Shear-dependent deposition and cluster growth mechanisms have been observed in tubular microreactors [13], while hydrodynamic modulation and oscillatory flow have been shown to delay encrustation in antisolvent crystallisation systems [14,15]. Moreover, kinetic analyses have established critical blockage criteria in reaction-driven crystallisation processes [16].

In continuous antisolvent precipitation, managing key process conditions, such as the antisolvent-to-solvent volumetric flow rate ratio (ASR) and the total flow rate, poses a trade-off between achieving small

* Corresponding authors.

E-mail addresses: f.galvanin@ucl.ac.uk (F. Galvanin), l.mazzei@ucl.ac.uk (L. Mazzei).

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particles at high throughput and limiting collision-induced aggregation and deposition. Nanoparticle production typically requires additives to stabilise nascent particles, making additive concentration an additional critical variable.

Despite the important role of process conditions, most fouling prevention strategies fail to address these trade-offs proactively, relying instead on reactive measures such as physical cleaning, chemical additives, and operational shutdowns to remove fouling deposits after they form [17,18]. Although effective in the short term, these methods introduce inefficiencies, reduce productivity, and cause product inconsistencies. Proactive strategies aim to optimise process parameters such as supersaturation, solvent selection, and stabilising agents, but they typically rely on empirical tuning. This limits their applicability across multivariate design spaces and their responsiveness to real-time fouling risks [18–20].

In continuous precipitation, real-time monitoring and control are essential for process stability. Process Analytical Technology (PAT) tools, including inline spectroscopy, have been used to monitor nucleation dynamics and manage supersaturation conditions [21–23]. However, these approaches are rarely applied to early-stage fouling detection or integrated with predictive or adaptive control frameworks. Beyond spectroscopy-based PAT, recent advances in probabilistic process monitoring have extended classical PCA/PLS approaches towards hybrid-variable and unsupervised detection frameworks for complex industrial data structures [24]. Improvements in precision flow diagnostics have similarly enhanced real-time hydrodynamic assessment in continuous systems [25]. However, these developments focus primarily on anomaly detection or measurement accuracy rather than adaptive identification of fouling-free operating regions.

Autonomous, closed-loop experimentation platforms have recently emerged in materials and molecular discovery, where Bayesian optimisation and experiment-theory feedback loops accelerate exploration of complex design spaces [26–28]; however, their application to fouling-driven instability in continuous precipitation systems remains largely unexplored. As pharmaceutical manufacturing shifts towards autonomous systems, there is a growing need for closed-loop approaches that combine real-time fouling detection with intelligent experimental design [9]. This study addresses that need by developing a model-guided, adaptive screening framework to identify feasible, non-fouling operating regions in continuous antisolvent precipitation.

Defining non-fouling, feasible operating conditions requires efficient exploration of the design space. Traditional approaches such as design of experiments (DoE) rely on fixed sampling plans, which can become inefficient when boundary regions are complex or poorly defined. Refining such boundaries often requires significantly more experiments, many of which may yield limited new information. To address this, we propose a closed-loop, model-guided screening framework [9] that combines an initial DoE with class balancing and active learning (AL) experimental design methods. The initial DoE generates feasibility labels (fouling or non-fouling), and class balancing targets underrepresented classes to produce a dataset that is both suitable for classification and adaptable to new fouling patterns encountered during experimentation [29,30]. A Gaussian process (GP) classifier provides a probabilistic framework for uncertainty estimation and is trained on these data to predict whether a given set of process conditions will lead to fouling [9]. AL iteratively selects the most informative experiments to improve the GP classifier model confidence [31,32]. This process continues until a user-defined experimental budget is reached or the classification uncertainty drops below a user-defined threshold. By integrating real-time learning into decision making, the proposed framework enables systematic and data-efficient characterisation of feasible operating regions in continuous antisolvent precipitation while supporting self-driven experimentation with minimal human intervention. Establishing fouling-free operating windows at an early development stage reduces empirical optimisation effort

and mitigates the risk of unexpected reactor instability during scale-up. This directly supports more robust and efficient implementation of continuous precipitation processes in pharmaceutical manufacturing.

In this work, fouling was detected using inline pressure sensors installed on both inlet lines of the reactor. Crucially, the pressure-based criterion flags the onset of fouling before physical blockage occurs, preventing clogging and preserving autonomous operation. A sustained rise in pressure at either inlet indicates increasing flow resistance due to fouling, allowing the system to autonomously classify the corresponding process conditions as infeasible. These classifications were used to train the GP model, which predicts fouling behaviour across the design space. This capability is integrated into an automated experimental platform [5] equipped with a graphical user interface (GUI) for controlling hardware components, enabling closed-loop operation.

The objectives of this study are to develop a closed-loop, autonomous exploration framework that uses real-time pressure feedback to map fouling and non-fouling operating regions; to apply a class balancing experimental design that ensures representative sampling of both classes; to construct a GP classification model with uncertainty quantification; to integrate AL for iterative selection of informative experiments; and to experimentally validate the approach for continuous ketoprofen precipitation in a CIJR. This work advances autonomous pharmaceutical manufacturing by establishing a robust, data-driven framework for closed-loop identification of non-fouling feasible operating conditions, improving process reliability and enabling self-optimising continuous precipitation systems.

2. Materials and experimental setup

2.1. Materials

The model drug used in this study was ketoprofen, a nonsteroidal anti-inflammatory drug (NSAID) known for its analgesic and anti-inflammatory properties. It is commonly used to treat pain, inflammation, and fever. Ketoprofen exhibits low aqueous solubility but dissolves more readily in organic solvents, a characteristic that influences its pharmaceutical formulation. Its poor water solubility motivates the use of nanoparticles, which increase specific surface area and dissolution rate, thereby enhancing bioavailability. Ketoprofen with a purity of $\geq 98.0\%$ was sourced from Fluorochem Limited.

De-ionised (DI) water ($\approx 14.5 \text{ M}\Omega$) was used as the antisolvent. Kollidon VA 64 (BASF), a copolymer of polyvinylpyrrolidone and vinyl acetate (PVP/VA), was employed as a matrix former, i.e., an additive that stabilises the drug in a solid dispersion. It improves solubility, and enhances drug bioavailability. The polymer was introduced into the system with the antisolvent phase. A 10 w/v% stock solution of PVP/VA in de-ionised water was prepared and stirred overnight to ensure complete dissolution. The additive concentration was varied by adjusting the flow rates of the antisolvent and the PVP/VA stock solution. $\geq 99.8\%$ ethanol (Fischer Scientific) was used as the solvent. Ketoprofen has a solubility in ethanol estimated at $\approx 47 \text{ w/v}\%$ [33]. The ketoprofen concentration in the solvent was 8.5 w/v%. For use as a cleaning solution, ethanol was diluted to 70% v/v with de-ionised water. All solutions were filtered through 0.22 μm nylon filter paper using vacuum filtration to remove dust and impurities that could act as unwanted nucleation sites, potentially leading to uncontrolled precipitation.

2.2. Reactor description

The reactor is a Confined Impinging Jet Reactor (CIJR) featuring a small, enclosed chamber, as shown in Fig. 1. The CIJR was 3D-printed using a Form 3 3D printer (Formlabs) and a high temperature HTV2 resin (Formlabs). 0.2 mm inlet nozzles were fabricated using a microdrill to drill perpendicularly to the length of the chamber, through the entire body of the reactor. This was done to ensure perfect

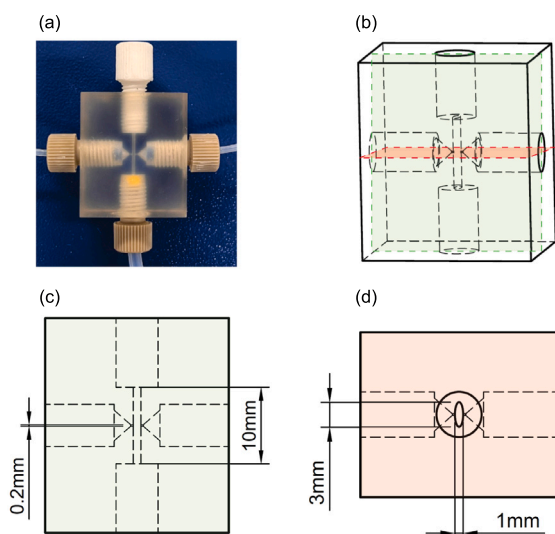


Fig. 1. Schematic representation of the CIJR used for continuous antisolvent precipitation. (a) Photograph of the 3D-printed CIJR. (b) Three-dimensional schematic of the reactor showing inlet nozzles and outlet configuration. (c) Longitudinal view along the chamber length showing the two opposing 0.2 mm inlet nozzles and the 10 mm mixing chamber length. (d) Transverse view of the mixing chamber showing the elliptical cross-section (3 mm × 1 mm).

alignment for the reactor inlets. The confined chamber has a length of 1 cm and an elliptical cross-section with 3 mm and 1 mm as dimensions of the two axes, as illustrated in Figs. 1(c) and (d).

Additionally, four standard microfluidic connections (1/4"-28) were drilled into the reactor: two connections for the inlet capillaries and two at the ends of the confined chamber; one serving as the particle suspension outlet and the other as an optional inlet, which remained closed in this study. The approximate volume of the CIJR was 0.024 mL. To provide additional residence time and ensure complete precipitation before sample collection, an additional PTFE tubing segment of internal diameter 1.59 mm was attached at the outlet, providing an effective total reactor volume of 1 mL.

2.3. Experimental platform

The antisolvent precipitation experiments were conducted on a fully automated continuous precipitation platform, adapted from a system developed in our previous work [5]. The experimental setup illustrated in Fig. 2 utilised three Milligat HF (Global FIA) piston pumps to deliver (1) ethanolic ketoprofen solvent solution, (2) DI water antisolvent, and (3) PVP/VA additive solution in DI water to the CIJR. The pumps were calibrated gravimetrically prior to experimentation. Gravimetric measurements at a nominal dispensed volume of 12.5 μ L showed a relative standard deviation below 1% for all pumps, indicating high delivery precision.

Pressure sensors (Pressure Unit, Fluigent) were placed at the CIJR inlet tubings to monitor pressure changes indicative of fouling. The inline pressure sensors were calibrated to zero under atmospheric conditions prior to operation and report gauge pressure relative to ambient conditions. Pressure was continuously monitored and processed according to the fouling detection protocol described in Section 3.3. Pressure relief valves (IDEX Health & Science, 100 psi) protected the system from overpressure. Pneumatically actuated four-way Swagelok ball valves (Swagelok SPL43Y) allowed for automated cleaning cycles between experiments.

The platform was controlled through a PharmaMV-based (Perceptive Engineering) GUI, shown in Fig. 3. The interface allows real-time control of process variables, stock solution inputs, and pressure-based

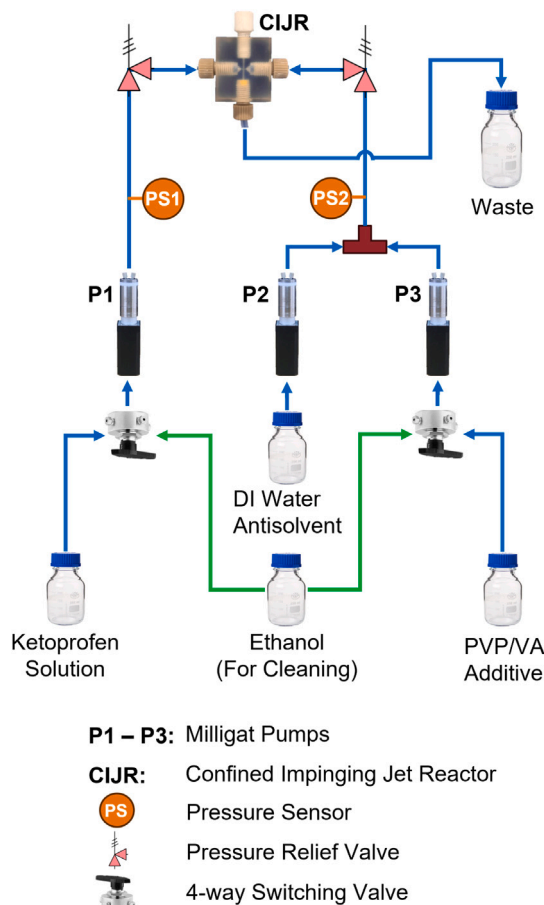


Fig. 2. Continuous-flow experimental setup for antisolvent precipitation of ketoprofen nanoparticles showing the CIJR, MilliGAT pumps (P1, P2 and P3) for solution delivery and inline pressure sensors (PS1 and PS2) to monitor inlet pressure for real-time fouling detection.

fouling detection, while also visualising live process data and algorithm status.

Instrument control and process automation were implemented using LabVIEW virtual instruments (VIs) for hardware communication, with a master VI coordinating system-wide operations. Data exchange between LabVIEW and PharmaMV was managed via OPC (Open Platform Communications) using network-published shared variables.

3. Methodology

This section presents our integrated approach for identifying non-fouling operating conditions in continuous antisolvent precipitation. The framework is formulated as a data-driven methodology that is not inherently restricted to a specific reactor geometry or solvent system. Provided that a measurable fouling indicator (e.g., pressure drop) and controllable process inputs are available, the same closed-loop strategy can be applied to different systems by retraining the classifier on system-specific data. We first establish the criterion for detecting fouling, then describe the experimental campaign, protocol and operating procedures, followed by the computational framework that guides the autonomous exploration of the design space.

3.1. Definition of the fouling criterion

To classify experiments as fouling (infeasible) or non-fouling (feasible), a pressure-based fouling criterion was established using a simplified reference setup that excluded the reactor and inlet tubing,

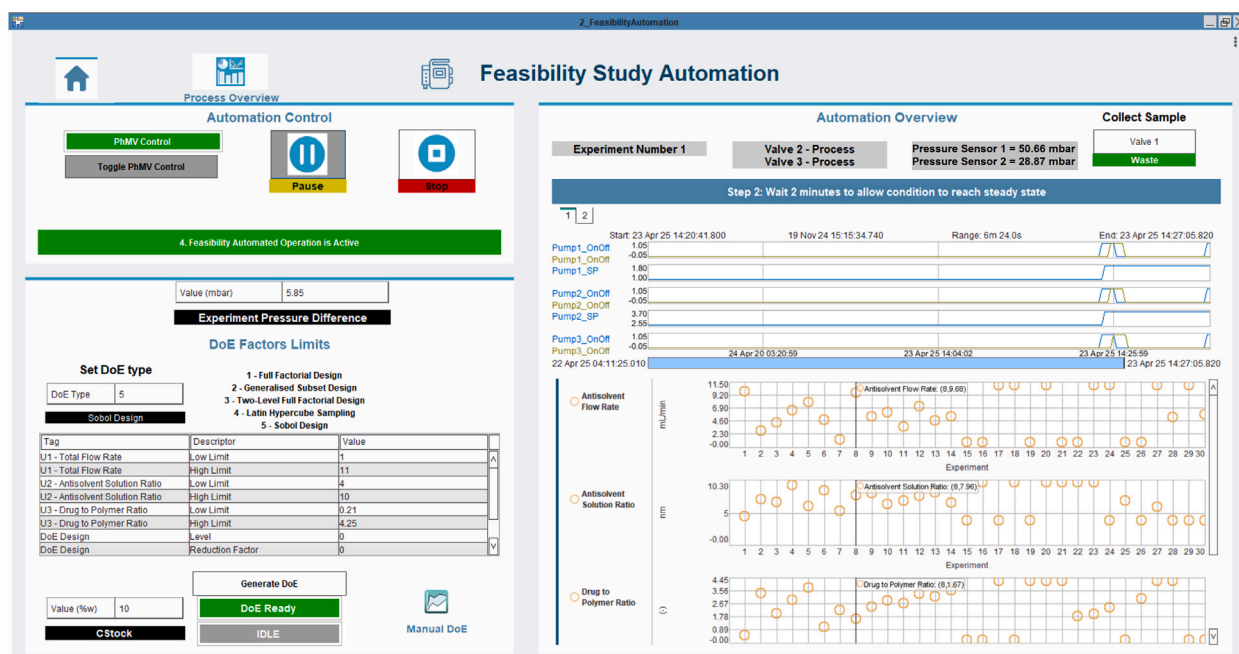


Fig. 3. PharmaMV graphical user interface (GUI) for the closed-loop autonomous experimental platform. The interface displays real-time pressure readings and experiment status and enables real-time monitoring and control of the precipitation process while implementing adaptive experimental design based on fouling detection feedback.

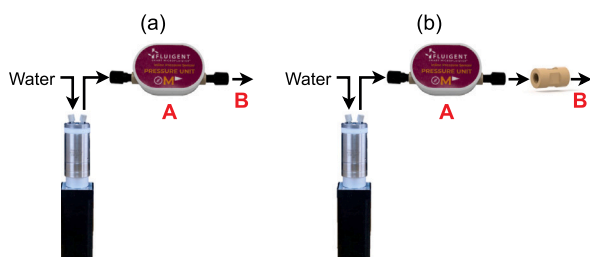


Fig. 4. Reference pressure-measurement setups used to define the fouling threshold. (a) Normal flow configuration: Point A is the gauge pressure sensor at the inlet of the outlet tubing; Point B is the outlet vented to atmosphere. (b) Localised constriction introduced between A and B using a precision union of known internal diameter. In both cases, the sensor reading equals $\Delta P_{\text{outlet}} = P_A - P_B$.

involving only the outlet tubing of the reactor. This configuration was based on the assumption that fouling predominantly occurs in the outlet region due to particle accumulation. This assumption was supported by preliminary experiments, which consistently indicated that clogging originated in the outlet region (see Supplementary Information, Section S1).

Fig. 4 illustrates the reference setup used to define the outlet pressure drop. A Milligat HF (Global FIA) piston pump was used to pump fluid through the outlet tubing. A pressure sensor (Pressure Unit, Fluigent) was installed at Point A, the inlet of the outlet tubing; the outlet at Point B was open to the atmosphere. The sensor was zeroed at atmospheric pressure so that its reading corresponded directly to the pressure drop across the outlet tubing:

$$\Delta P_{\text{outlet}} = P_A - P_B, \quad \text{with } P_B = P_{\text{atm}} = 0 \text{ gauge} \quad (1)$$

To establish a fouling threshold, we measured ΔP_{outlet} under two conditions: (i) normal, unconstricted flow and (ii) constricted flow, where a precision union of known internal diameter was inserted to represent a localised narrowing of the flow path. This approach avoids deliberate fouling with API and instead provides a controlled, repeatable pressure drop increase consistent with outlet fouling. This

controlled constriction is not intended to replicate the physicochemical mechanisms of deposit formation, but rather to establish a reproducible hydraulic threshold corresponding to an operationally unacceptable level of flow restriction.

The two reference pressure drops are denoted as $\Delta P_{1,\text{ref}}$, corresponding to the pressure drop under normal, unconstricted flow, and $\Delta P_{2,\text{ref}}$, corresponding to the pressure drop with a known constriction inserted at the outlet.

Fig. 4a corresponds to $\Delta P_{1,\text{ref}}$, while the setup in Fig. 4b includes the constriction and is used to determine $\Delta P_{2,\text{ref}}$. The reference pressure drop threshold ($\Delta P_{\text{th,ref}}$) is defined as the difference between the two:

$$\Delta P_{\text{th,ref}} \equiv \Delta P_{2,\text{ref}} - \Delta P_{1,\text{ref}} \quad (2)$$

Under laminar conditions, the pressure drop depends on the flow rate and the fluid viscosity. All reference measurements were performed at a reference flow rate $Q_{\text{ref}} = 1 \text{ mL min}^{-1}$ using DI water, which has low, well-characterised viscosity. Although the additive-containing antisolvent stream used during experiments is more viscous, its viscosity is of the same order of magnitude, making DI water a conservative but appropriate reference fluid. The outlet tubing used for all measurements had an inner diameter of 1.59 mm. For the constricted case, precision unions with known internal diameters (IDEX Health & Science, P-703, P-710, P-702) were inserted to represent different levels of localised tubing narrowing. From the measured pressure drops, a calibration table (Table 1) was constructed to interpolate between pressure drops corresponding to different constriction levels. In this study, a 30% reduction in tubing diameter (from 1.59 mm to 1.11 mm) was chosen as the fouling threshold. This value was selected as a representative and practically significant reduction in effective flow area that produces a clearly observable increase in pressure drop. It is not derived from a specific fouling correlation, but instead defines a user-selected operational instability criterion. The framework allows this threshold to be adjusted depending on the desired level of conservatism without altering the underlying methodology. For this level of constriction:

$$\begin{aligned} \Delta P_{1,\text{ref}} &= 44.0 \text{ mbar}, \quad \Delta P_{2,\text{ref}} = 49.9 \text{ mbar} \\ \Rightarrow \Delta P_{\text{th,ref}} &= 5.9 \text{ mbar} \end{aligned} \quad (3)$$

Table 1

Outlet pressure drop at $Q_{\text{ref}} = 1 \text{ mL min}^{-1}$ (DI water) for the baseline tubing and for precision unions inserted between Point A (sensor) and Point B (atmosphere). Internal diameter refers to the flow passage: the outlet tubing for the baseline row and the precision-union passage for the other rows.

Case	Internal diameter (mm)	Pressure drop (mbar)
Baseline (tubing)	1.59	44.0
P-703 ($\approx 20\%$ constriction)	1.25	48.8
P-710 ($\approx 50\%$ constriction)	0.75	53.0
P-702 ($\approx 70\%$ constriction)	0.50	57.9

Since in laminar conditions the pressure drop scales linearly with the flow rate, the threshold was scaled proportionally to the total flow rate Q used in each experiment, that is:

$$\Delta P_{\text{th}}(Q) = \frac{Q}{Q_{\text{ref}}} \times \Delta P_{\text{th,ref}} \quad (4)$$

3.2. Application of the fouling threshold

During each experiment, gauge pressure was monitored at both inlets using inline sensors installed upstream of the reactor. These sensors were calibrated to zero under atmospheric conditions and reported the inlet pressure relative to atmosphere.

Fouling was detected by tracking changes in inlet pressure over time. Once steady-state flow was established, the baseline inlet pressure $P_{t=0}$ was recorded. Pressure P_t was then measured until $t = 10 \text{ min}$. A sustained pressure rise indicates increasing flow resistance due to potential fouling downstream.

A process condition was classified as fouling (infeasible) if the pressure increase at either inlet exceeds the fouling threshold $[\Delta P_{\text{th}}(Q)]$ derived from controlled outlet constriction tests:

$$P_t - P_{t=0} > \Delta P_{\text{th}}(Q) \implies \text{Fouling} \quad (5)$$

This classification was performed automatically and in real time during each run.

3.3. Experimental campaign and protocol

The experimental campaign followed an autonomous algorithm implemented through custom Python scripts, which were integrated into the PharmaMV–LabVIEW environment. The approach began with a DoE to initiate model training, followed by a class balancing routine to ensure adequate representation of both fouling and non-fouling conditions. A GP classifier then guided an AL module that selected the next experiment to reduce model uncertainty. This process continued until either a user-defined experimental budget of 50 experiments was reached or the classification uncertainty of the model dropped below a threshold, as defined in Section 3.8. This closed-loop strategy enabled real-time, operator-free decision making during automated runs, allowing the system to efficiently map the boundary between feasible and infeasible conditions with minimal experimental burden. The overall workflow is illustrated in Fig. 5.

For each experiment in the campaign, the following protocol was implemented:

1. System inputs (total flow rate, ASR and drug-to-additive mass ratio) were set according to the computational design from the automation algorithm.
2. Pump flow rates were calculated based on the system inputs.
3. The pressure threshold was scaled proportionally to the total flow rate for each experiment using Eq. (4).

4. After allowing three residence times through the reactor and outlet tubing, we assumed steady state ($t = 0$). Online pressure measurements were then recorded for 10 min. This time frame was deemed appropriate for detecting meaningful fouling while allowing for potential self-correction of transient pressure fluctuations.
5. Pressure data were recorded continuously, including during the residence-time waiting period, at 0.01 s intervals. As this high acquisition rate results in occasional random spikes or transient fluctuations, a moving average filter was applied so that the reported pressure at time t , denoted as P_t , represents a locally averaged value rather than the instantaneous measurement. The discrete-time moving average used in this work is defined as:

$$P_t[n] \equiv \frac{1}{M} \sum_{k=0}^{M-1} P_{\text{inst}}[n-k] \quad (6)$$

where $P_{\text{inst}}[n]$ is the instantaneous pressure sample, $P_t[n]$ is the filtered pressure value used for fouling assessment, M is the window size, n is the discrete-time sample index, and k is the summation index. A window size of $M = 100$ samples was used. As pressure data were recorded during the three-residence-time period prior to defining $t = 0$, at least M samples were already available when evaluating Eq. (6), ensuring that all terms in the summation were defined. The resulting smoothed pressure trace was then used to calculate the pressure evolution between $P_{t=0}$ and $P_{t=10}$.

6. The experiment was then classified as fouling or non-fouling as per Eq. (5).
7. A cleaning cycle was then initiated to restore baseline pressure conditions prior to the next experiment. During cleaning, the pneumatically actuated four-way valves switched the flow path from the reactant streams to the cleaning solution (70% v/v ethanol in deionised water). Pumps 1 and 3 were operated at 2.5 mL min^{-1} to flush the reactor and outlet tubing for two minutes, which was sufficient to restore baseline pressure conditions before the next experiment.

3.4. Treating fouling as a classification problem

Let us consider an experiment identified by a set of input variables lumped in the N_u -dimensional vector $\mathbf{u}$. When we run such an experiment to study the fouling in the continuous flow crystallisation process, the experiment can have either a positive outcome ($c = 1$ or *non-fouling*) or a negative outcome ($c = 0$ or *fouling*). In theory, this experiment is deterministic, resulting either in $c = 1$ or in $c = 0$, depending on the conditions and how they are affecting the underlying fouling mechanisms. However, because the detailed mechanisms governing fouling are often complex, uncertain, or partially unobservable, it is convenient to represent fouling in probabilistic terms. In this framework, we assume that the outcome c of an experiment $\mathbf{u}$ is random. This means that once we select an experiment identified by the input vector $\mathbf{u}$, the outcome can be both $c = 1$ and $c = 0$, but with associated probabilities. The probability of $c = 1$ is ϕ , which is a function of $\mathbf{u}$, and the probability of $c = 0$ is $1 - \phi$. Consequently, being c a discrete random variable, we associate with c a probability mass function (PMF), which is a Bernoulli distribution [34] because c is a binary random variable. This PMF is:

$$f[c; \phi(\mathbf{u})] = \begin{cases} \phi(\mathbf{u}), & \text{for } c = 1 \\ 1 - \phi(\mathbf{u}), & \text{for } c = 0 \end{cases} \quad (7)$$

where $\phi(\mathbf{u})$ denotes the probability of occurrence of the positive outcome ($c = 1$). As f depends on the variable c and features the parameter $\phi(\mathbf{u})$, we write $f[c; \phi(\mathbf{u})]$. Eq. (7) can be equivalently written as:

$$f[c; \phi(\mathbf{u})] = [\phi(\mathbf{u})]^c [1 - \phi(\mathbf{u})]^{1-c} \quad \text{with } c \in \{0, 1\} \quad (8)$$

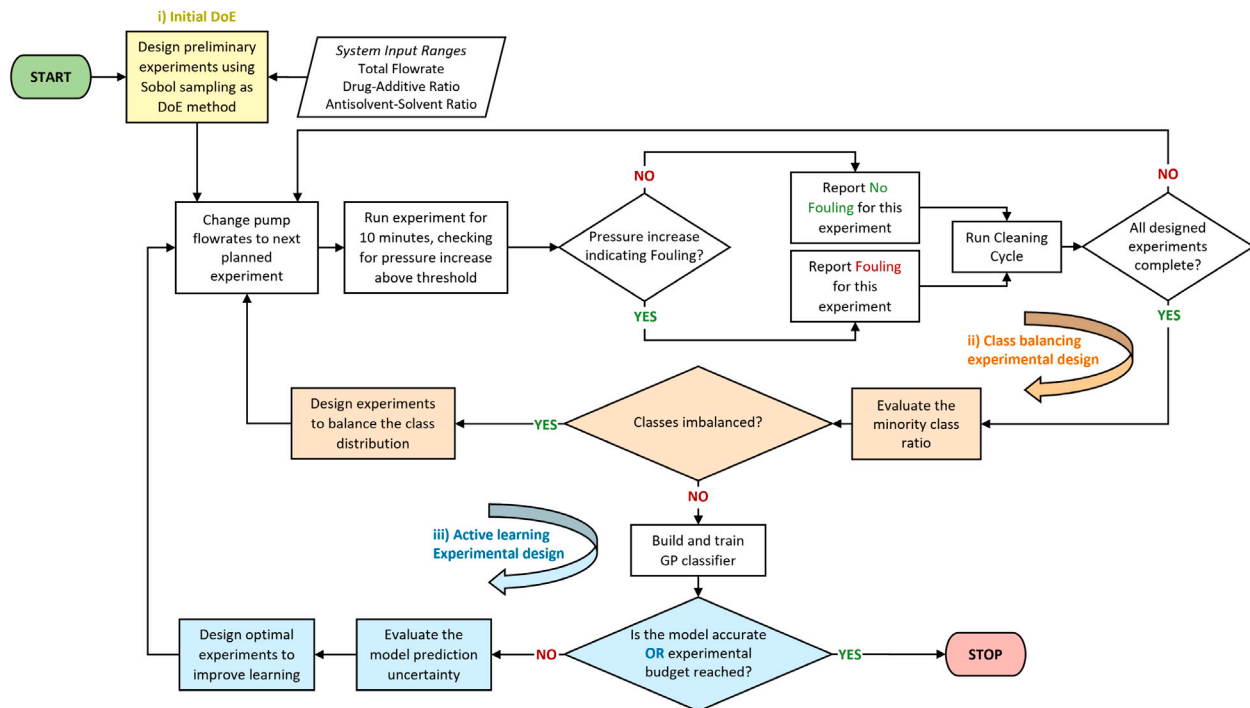


Fig. 5. Autonomous screening algorithm flowchart illustrating the closed-loop approach for identifying non-fouling operating conditions. The process integrates (i) initial DoE experimentation, (ii) class balancing to improve dataset representation, and (iii) AL with GP classification to iteratively select the most informative experiments. The loop continues until a predefined experimental budget is reached or model accuracy criteria are satisfied.

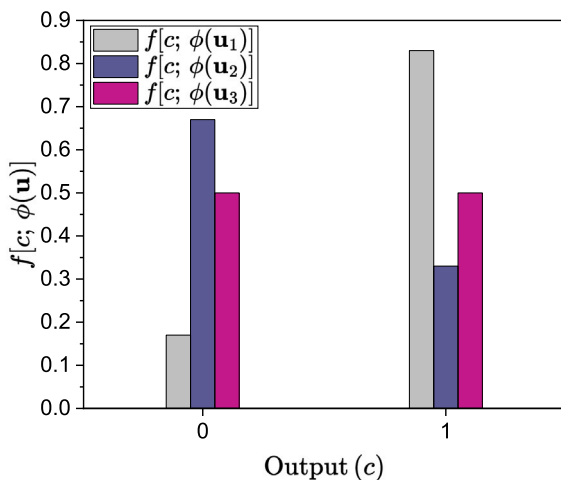


Fig. 6. Three examples of Bernoulli distribution used to represent the chances of fouling at three experiments identified by the input vectors $\mathbf{u}_1$, $\mathbf{u}_2$ and $\mathbf{u}_3$.

An illustration of the random behaviour of experiments under the probabilistic framework is provided in Fig. 6. As shown in the figure, corresponding to the three experiments identified by the input vectors $\mathbf{u}_1$, $\mathbf{u}_2$ and $\mathbf{u}_3$, the probabilities of successes ($c = 1$ or non-fouling) are indicated by the corresponding PMFs. In this probabilistic framework, to classify experiments as fouling and non-fouling, we introduce a threshold value ϕ_c and regard an experiment as successful or non-fouling if its probability of success is greater than ϕ_c . Mathematically, this can be written as:

$$\text{Experiment } \mathbf{u} \Rightarrow \begin{cases} \phi(\mathbf{u}) > \phi_c, & \text{non-fouling} \\ \phi(\mathbf{u}) \leq \phi_c, & \text{fouling} \end{cases} \quad (9)$$

The value of ϕ_c is arbitrary and is decided based on the purpose of classification. The methodology used to obtain the optimal value for ϕ_c is discussed later.

3.5. Binary classification using GPs

The essence of developing a binary classification model in a probabilistic framework involves modelling the relationship between $\mathbf{u}$ and ϕ . To model this relationship, we employed GPs for several reasons. For the small, low-dimensional datasets typically available from experimental platforms, GPs are particularly well suited compared with tree-based and artificial neural network (ANN) models. They can be effectively trained with as few as 20–100 data points, generally providing accurate interpolation. Moreover, GPs naturally incorporate prior information through the choice of kernel functions and through priors placed on kernel hyperparameters. For instance, if prior knowledge suggests that the process behaviour (here, fouling behaviour) is smooth, this assumption can be directly encoded via an appropriate kernel choice. In addition, GPs provide principled, built-in uncertainty quantification, which is essential for the design of closed-loop, automated experiments aimed at developing predictive models. In contrast, tree-based and ANN models do not directly yield reliable uncertainty estimates. Finally, GP models typically involve fewer hyperparameters than these alternatives, making their predictions less sensitive to hyperparameter tuning. This is a significant advantage when working with small datasets, for which the optimisation of a large number of hyperparameters is particularly challenging.

As stated, $\phi(\mathbf{u})$ is the probability of occurrence of the positive outcome (non-fouling experiment here) for the experiment identified by $\mathbf{u}$. In the GP framework, this probability itself is treated as a random variable; i.e., one is not certain about its value. Since ϕ is a continuous random variable, one can associate with ϕ a probability density function (PDF). If r is the PDF associated with ϕ , while ψ is the sample-space variable corresponding to the random variable ϕ , then $r(\psi; \mathbf{u})d\psi$ represents the probability that ϕ lies in the range $d\psi$ around

the value ψ for the experiment identified by the vector $\mathbf{u}$. Since the PDF of ϕ is parametric on the continuous variable $\mathbf{u}$, ϕ is referred to as a random process.

As the continuous random variable ϕ can only vary between zero and unity, its PDF cannot be Gaussian. However, GP modelling is built on the assumption that the random variable is represented by a Gaussian distribution. Hence, within a standard GP framework, one cannot directly work in terms of the variable ϕ . Instead, one introduces a random variable h , which ranges over the real line and whose PDF is assumed to be Gaussian. This random variable is called a *latent variable*. ϕ and h are related via a nonlinear transformation, such as the sigmoid transformation:

$$\phi(h) = \frac{1}{1 + e^{-h}} \quad (10)$$

We selected the sigmoid transformation for its smoothness (being continuous and differentiable) and analytical simplicity (easy to compute with a closed-form derivative), which, as explained later, are important properties for the AL method employed in this study. Similarly to ϕ , also h is a random process, because its PDF is parametric on the continuous variable $\mathbf{u}$.

The latent variable h at a particular experiment $\mathbf{u}_1$ is characterised by its Gaussian PDF, which is concisely represented as:

$$[h_1; \mathbf{u}_1] \sim \mathcal{N}[m(\mathbf{u}_1), k(\mathbf{u}_1, \mathbf{u}_1)] \quad (11)$$

where h_1 denotes the sample-space variable corresponding to the random variable $h(\mathbf{u}_1)$ such that $p(h_1; \mathbf{u}_1)dh_1$ is the probability that h lies in the range dh_1 around h_1 for the experiment $\mathbf{u} = \mathbf{u}_1$. In Eq. (11) and henceforth, the symbol $\sim$ means 'is distributed as' and $\mathcal{N}[m(\mathbf{u}_1), k(\mathbf{u}_1, \mathbf{u}_1)]$ denotes a Gaussian (or normal) distribution with mean $m(\mathbf{u}_1)$ and variance $k(\mathbf{u}_1, \mathbf{u}_1)$. The GP $h(\mathbf{u})$ implies that any finite-dimensional marginal such as the joint distribution of $h(\mathbf{u}_1)$ and $h(\mathbf{u}_2)$ is multivariate Gaussian. Thus, the 2-point marginal at $\mathbf{u}_1$ and $\mathbf{u}_2$ is described by a bivariate normal PDF:

$$[h_1, h_2; \mathbf{u}_1, \mathbf{u}_2] \sim \mathcal{N}\left\{ \begin{bmatrix} m(\mathbf{u}_1) \\ m(\mathbf{u}_2) \end{bmatrix}, \begin{bmatrix} k(\mathbf{u}_1, \mathbf{u}_1) & k(\mathbf{u}_1, \mathbf{u}_2) \\ k(\mathbf{u}_2, \mathbf{u}_1) & k(\mathbf{u}_2, \mathbf{u}_2) \end{bmatrix} \right\} \quad (12)$$

where h_1 and h_2 are the two sample-space variables of h associated with the experiments $\mathbf{u}_1$ and $\mathbf{u}_2$ such that $p(h_1, h_2; \mathbf{u}_1, \mathbf{u}_2)dh_1dh_2$ is the probability that h lies in the range dh_1 around h_1 at $\mathbf{u} = \mathbf{u}_1$ and at the same time in the range dh_2 around h_2 at $\mathbf{u} = \mathbf{u}_2$. A graphical representation of the marginal PDFs of h at two experiments $\mathbf{u}_1$ and $\mathbf{u}_2$ is provided in Fig. 7. Note that the marginal PDFs are plotted for one of the input variables, say u_1 . Based on Fig. 7, imagine the joint PDF of h for two values of $\mathbf{u}$, $\mathbf{u}_1$ and $\mathbf{u}_2$. This is also the 2-point marginal of h at $\mathbf{u}_1$ and $\mathbf{u}_2$. This 2-point marginal PDF can be defined using Eq. (12), which is also annotated inside Fig. 7. By extending this definition, the values of the latent variable h at any finite set of experiments denoted by the vectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ can be concisely represented by the multivariate Gaussian PDF:

$$[h_1, \dots, h_n; \mathbf{u}_1, \dots, \mathbf{u}_n] \sim \mathcal{N}(\mathbf{m}, \mathbf{K}) \quad (13)$$

where $h_1, \dots, h_n$ are the n sample-space variables, while $\mathbf{u}_1, \dots, \mathbf{u}_n$ are the n parameters, $[\mathbf{m}]_i = m(\mathbf{u}_i)$ is the mean and $[\mathbf{K}]_{ij} = k(\mathbf{u}_i, \mathbf{u}_j)$ is the covariance. In the binary classification using GPs, we assume that the latent function $h(\mathbf{u})$, $\mathbf{u} \in \mathcal{U}$, is a random process, specifically a *Gaussian process* such that for any finite set of experiments $\mathbf{u}_1, \dots, \mathbf{u}_n \in \mathcal{U}$, the random vector $[h_1, \dots, h_n]^T$ has a multivariate normal distribution.

The latent function $h(\mathbf{u})$ can be completely specified by the GP with a mean function $m(\cdot)$ and a covariance or kernel function $k(\cdot, \cdot)$. This mean function and kernel function are functions of $\mathbf{u}$, because the GP is parametric in $\mathbf{u}$. For any two input vectors $\mathbf{u}$ and $\mathbf{u}'$, the covariance between the corresponding sample-space variables h and h' is defined as:

$$\text{Cov}(h, h'; \mathbf{u}, \mathbf{u}') = k(\mathbf{u}, \mathbf{u}') \quad (14)$$

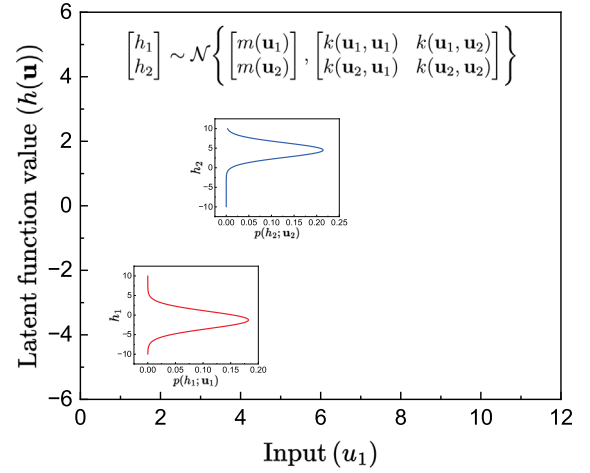


Fig. 7. Marginal PDFs of the GP latent function $h(\mathbf{u})$ for two experiments, $\mathbf{u}_1$ and $\mathbf{u}_2$.

Eq. (14) signifies that the covariance between the latent function values is expressed as a function of the corresponding input vectors. The covariance function $k(\cdot, \cdot)$ is also a parametric function of hyperparameters θ , which are optimised during the model training. In this work, as covariance function, we have employed a Matérn-5/2 kernel with trainable hyperparameters $\theta = \{\sigma_h^2, l\}$, representing the signal variance and the length scale [35–37].

Once the GP is known, the PDF of the latent variable h is known and then the PDF of ϕ can be calculated using the nonlinear transformation [Eq. (10)] relating h to ϕ . Then to classify an experiment as fouling or non-fouling, we consider the mean value of ϕ and compare it with the threshold value ϕ_c . Therefore, we can modify Eq. (9) to write the decision rule as:

$$\text{Experiment } \mathbf{u} \Rightarrow \begin{cases} \langle \phi(\mathbf{u}) \rangle > \phi_c, & \text{non-fouling} \\ \langle \phi(\mathbf{u}) \rangle \leq \phi_c, & \text{fouling} \end{cases} \quad (15)$$

where the symbol $\langle \cdot \rangle$ denotes the expectation. However, this classification is uncertain, its uncertainty being quantified by the variance of the PDF of ϕ . The GP model is valuable insofar as it provides both a prediction of an experiment's success and a measure of the associated classification uncertainty.

The training of the classifier model is based on Bayes' theorem applied to the latent function values. Consider a training dataset of n observations, where the inputs vector for all the n cases are aggregated in the $n \times N_u$ matrix $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_n]^T$, whereas the corresponding class labels are collected in the vector $\mathbf{c} = [c_1, \dots, c_n]^T$. The posterior $p(\mathbf{h} | \mathbf{U}, \mathbf{c}, \theta)$ over the latent function values $\mathbf{h} = [h_1, \dots, h_n]^T$, given the hyperparameters θ is evaluated using the Bayes' theorem:

$$p(\mathbf{h} | \mathbf{U}, \mathbf{c}, \theta) = \frac{p(\mathbf{c} | \mathbf{h}, \mathbf{U}, \theta) p(\mathbf{h} | \mathbf{U}, \theta)}{p(\mathbf{c} | \mathbf{U}, \theta)} \quad (16)$$

The classifier model training is carried out by maximising the log marginal likelihood:

$$\hat{\theta} = \arg \max_{\theta} \log p(\mathbf{c} | \mathbf{U}, \theta) \quad (17)$$

This computation is intractable. An approximate solution to Eq. (17) is obtained by maximising the variational inference [38] loss function — evidence lower bound (ELBO) [39]. After model training, the predictions for a new point $\mathbf{u}_*$ is:

$$p(c_* = 1 | \mathbf{u}_*, \mathbf{U}, \mathbf{c}) = \int \phi[h(\mathbf{u}_*)] p(h(\mathbf{u}_*) | \mathbf{u}_*, \mathbf{U}, \mathbf{c}) dh(\mathbf{u}_*) \quad (18)$$

In this work, the GP model development and training were implemented using the Pyro [40] probabilistic programming framework,

with the Adam optimiser [41] from the PyTorch library employed for optimising the model parameters during training.

3.6. DoE for generating initial data

The initial phase of our autonomous screening approach used DoE methods to provide a starting point for the GP model training. The first eight experiments formed the DoE stage, where the experimental conditions were chosen using Sobol sampling [42,43]. This sampling was selected to capture the potential nonlinear relationship between the inputs $\mathbf{u}$ and the binary variable c , which is also known as the class variable, using minimum data points or experiments. The goal of this stage was to explore the design space and generate an initial classification dataset that reflects the overall behaviour of the system.

The DoE explored three critical process variables within defined operating ranges: total flow rate (1–11 mL min⁻¹), constrained by the operational limitations of the milliGAT pumps; antisolvent-to-solvent volume flow rate ratio (ASR, 4–10 mL min⁻¹/mL min⁻¹), spanning solvent-rich to antisolvent-rich feeds and thereby modulating supersaturation for ketoprofen; and drug-to-additive mass ratio (0.2125–4.25 g/g), selected to ensure additive concentrations remained within practical limits (0.2% to below the 10% stock solution concentration) while accommodating the selected ASR range.

3.7. Class balancing experimental design

The primary objective of the DoE was to explore the design space and generate a preliminary dataset to train the classifier model. However, the Sobol sampling approach or, in general, any other DoE methods do not guarantee a balanced distribution of experimental outcomes across feasible and infeasible conditions or across non-fouling and fouling conditions. The ability of a binary classifier model to determine the accurate decision boundary that separates classes, such as “fouling” and “non-fouling”, depends on the distribution of the data used to train it [44,45]. When the dataset is highly imbalanced, with one class being substantially overrepresented, the classifier lacks sufficient information about the minority class. Under such circumstances, it is important to either generate synthetic minority data points using data augmentation methods [46] or design experiments that can generate data points belonging to the minority class.

Algorithm 1: Class balancing experimental design

Input: $\mathbf{U}, \mathbf{c}$
 1 /* $\mathbf{U}$ is the input array of size $n \times N_u$ and $\mathbf{c}$ is the output array of size $n \times 1$. Corresponding to each row $\mathbf{u}_i$ of $\mathbf{U}$, there is an output value c_i , where $c_i \in \{0, 1\}$. */
Output: $\mathbf{U}_d$
 2 /* $\mathbf{U}_d$ is the $d \times N_u$ dimensional input array which suggests the conditions for the next d experiments. */
 3 Evaluate $\mathbf{U}_0$ such that $\mathbf{U}_0 \in \mathbf{U}$ and all the rows of $\mathbf{U}_0$ result in $c = 0$. /* Infeasible experiments */
 4 Evaluate $\mathbf{U}_1$ such that $\mathbf{U}_1 \in \mathbf{U}$ and all the rows of $\mathbf{U}_1$ result in $c = 1$. /* Feasible experiments */
 5 Select the minority sample $\mathbf{u}_0$, which is the nearest minority point to the midpoint of the design space.
 6 Select N_u nearest neighbours of $\mathbf{u}_0$ (giving priority to minority points), where N_u is the dimension of the design space.
 7 Form a simplex geometry with $\mathbf{u}_0$ and its selected N_u neighbours as the vertices.
 8 Suggest the array of new designed experiments $\mathbf{U}_d$ such that it includes the midpoints of all the edges of the simplex, each of which has at least one minority vertex point.

In this work, we have developed a class balancing experimental design method that belongs to the latter category of approaches. Our

method is inspired by the Nelder–Mead simplex method [47] used in numerical optimisation and the synthetic minority oversampling technique (SMOTE) [46] used to balance the class distribution using the data augmentation method. Let n denote the number of performed experiments and N_u denote the number of input process variables (in our case equal to 3). Let $\mathbf{U}$ and $\mathbf{c}$ denote a finite set of data points with the input points $\mathbf{U} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n)$ and the vector of class labels $\mathbf{c} = [c_1, c_2, \dots, c_n]^T$. The i -th row of the $n \times N_u$ dimensional array $\mathbf{U}$ represents the conditions (total flow rate, ASR, drug-to-additive ratio) of the i -th experiment, defined by the vector $\mathbf{u}_i$. Similarly, the i -th row of the $n \times 1$ dimensional array $\mathbf{c}$ has a value of 0 (if the experiment caused fouling) or 1 (if the experiment did not cause fouling). Given $\mathbf{U}$ and $\mathbf{c}$, the class balancing experimental design method proposed in this work suggests the conditions of the next d experiments, denoted by $\mathbf{U}_d$. These experiments are designed to balance the distribution of the class labels such that after performing these experiments, we expect to get roughly equal numbers of zeros and ones within the updated $\mathbf{c}$. The algorithm of the proposed method is given in Algorithm 1. The method is applicable to problems with at least one minority data point.

The minority class ratio, defined as the ratio of the number of minority class instances to the total number of instances, was monitored after the DoE phase. The decision to invoke the class balancing algorithm was made autonomously based on the minority class ratio. A user-defined threshold of 0.4 was set: if the observed minority class ratio fell below this value, this indicated an imbalanced class distribution, and the class balancing experimental design was triggered to generate conditions favouring the minority class.

3.8. AL experiments

As stated, the trained GP model provides not only predictions of class labels but also an estimate of the uncertainty associated with those predictions. This uncertainty gives meaningful insight into the confidence of the model and helps guide decisions on what data to collect next. In this context, the next experiments are chosen in the regions of the operating space where the uncertainty of the model is largest, so that this uncertainty can be reduced. This approach is called AL. It is the field of machine learning (ML) that is analogous to optimal experimental design and leverages model uncertainty to identify and select informative training data to develop reliable ML models [28,48].

Unlike Bayesian optimisation (BO), which seeks to identify the optimum operating point (with respect to a cost or performance objective) as quickly as possible by optimising an acquisition function such as probability of improvement, expected improvement or upper confidence bound, active learning (AL) aims to develop an accurate (low prediction error) and reliable (low prediction uncertainty) model by improving the model training or learning process. In BO, new samples are selected at points where the promise of finding a better value is large, whereas in AL, new samples are picked by maximising the uncertainty of predictions. Thus, when optimisation is the goal, BO is the preferred approach; in contrast, when the goal is to develop a predictive model, AL is the preferred approach. Typically, BO is used with a regressor. Since the focus of this study is on the development of a predictive binary classification model to characterise fouling within an antisolvent precipitation process, AL is the suitable algorithm.

AL plays an important role in building reliable GP classification models using a small amount of data. Instead of randomly collecting data, AL focuses on selecting the most informative data points where the model behaviour is least confident (highly uncertain predictions). Because GP models can measure how uncertain they are about their predictions, this uncertainty can guide the selection of new data. As a result, even with limited data, a GP classifier can become accurate and reliable if the data are chosen wisely through AL.

Generally, the uncertainty in the GP classification is expressed as the variance of the posterior predictions of the latent function values. This variance at a new input point $\mathbf{u}_*$ can be computed as:

$$\sigma_h^2(\mathbf{u}_*) = k(\mathbf{u}_*, \mathbf{u}_*) - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_* \quad (19)$$

where $\mathbf{K}$ is the $n \times n$ covariance matrix with elements defined by $[\mathbf{K}]_{ij} = k(\mathbf{u}_i, \mathbf{u}_j)$, $i, j = 1, \dots, n$, and $\mathbf{k}_* = [k(\mathbf{u}_1, \mathbf{u}_*), \dots, k(\mathbf{u}_n, \mathbf{u}_*)]^\top$. As discussed in Pankajakshan et al. [49], Eq. (19) captures the uncertainty in predicting the latent function values, which is different from the uncertainty in the predicted probability values, which is ultimately used to decide the class labels. To evaluate the uncertainty in the predicted probability values, we utilised the propagated uncertainty AL objective function proposed in Pankajakshan et al. [49]; this was employed to design the experimental conditions to generate new training data for the classifier model. The AL problem was designed as the optimisation problem of the form:

$$\mathbf{u}_{\text{al}} = \arg \max_{\mathbf{u} \in U} \sigma_\phi^2(\mathbf{u}) \quad (20)$$

where the propagated uncertainty objective function $\sigma_\phi^2(\mathbf{u})$ is defined as:

$$\begin{aligned} \sigma_\phi^2(\mathbf{u}) &\equiv \left[\frac{d\phi}{dh} \Big|_{\mathbf{u}} \cdot \sigma_h(\mathbf{u}) \right]^2 = \left[\frac{e^h}{(1+e^h)^2} \right]^2 \cdot \sigma_h^2(\mathbf{u}) \\ &= \left\{ \phi(h)[1 - \phi(h)] \right\}^2 \cdot \sigma_h^2(\mathbf{u}) \end{aligned} \quad (21)$$

The choice of sigmoid transformation allowed us to easily evaluate the propagated uncertainty provided in Eq. (21) and effectively solve the optimisation problem of Eq. (20). Note that the propagated uncertainty AL objective function provided in Eq. (21) is based on a first-order (i.e., linear) approximation of the nonlinear transformation $\phi(h)$. The resulting variance in the transformed space becomes increasingly accurate as the latent uncertainty decreases with more training data.

For classification problems, existing objective functions for AL are based on some measure of the classification uncertainty. Popular choices are classification entropy, classification uncertainty calculated as the product of predicted class probabilities, and Bayesian active learning by disagreement (BALD). A limitation with both classification uncertainty and classification entropy is that they are based on the predicted class probability values without considering the uncertainty around these values. As a result, when using either classification uncertainty or classification entropy as the AL objective function, new samples will be suggested near the predicted boundary separating the classes, and the iterative learning of the true decision boundary would be slow. On the other hand, BALD is a very powerful AL objective function that selects new samples by maximising the marginal entropy in the predictive probability distribution, obtained by marginalising over the GP latent function, while prioritising regions where uncertainty arises from disagreement among plausible latent functions [50]. Our method – a propagated uncertainty AL objective function – is functionally similar to BALD, but instead of maximising marginal entropy in the probability space, it maximises the propagated uncertainty in the probability space. Although not tested against BALD, our method was found to outperform the widely used uncertainty sampling AL methods [49].

At each iteration of the autonomous platform, the GP classifier model was retrained using the entire dataset available at that stage to update its predictions and uncertainty estimates. The posterior propagated uncertainty $\sigma_\phi^2(\mathbf{u})$ across the design space was evaluated using Eq. (21). The next experimental condition was selected by solving the AL problem given by Eq. (20), thereby targeting the operating condition associated with maximum uncertainty. The AL optimisation problem in Eq. (20) was solved using the Adam optimiser [41] implemented in the PyTorch library [51]. This closed-loop operation continued iteratively, progressively refining the classifier model with each new experiment.

The closed-loop operation of the computational framework is terminated on either reaching the maximum experimental budget (50 experiments set in this study) or satisfying the threshold (i.e., $\leq$ the threshold) set on the largest value (the worst value or the least favourable value in terms of classification uncertainty) of the posterior propagated uncertainty. A threshold of 0.01 was set as the uncertainty threshold on the largest value (worst value) of the posterior propagated variance σ_ϕ^2 predicted across the design space.

3.9. Model performance evaluation

The trained GP classifier predicts the probability that an experiment belongs to the non-fouling class ($c = 1$). However, as stated in Sections 3.4 [Eq. (9)] and 3.5 [Eq. (15)], the final class assignment depends on the value of the classification threshold ϕ_c . For instance, if ϕ_c is set to 0.6, only experiments with predicted probabilities greater than 0.6 are classified as non-fouling ($c = 1$); otherwise, they are assigned to the fouling class ($c = 0$). In this work, the performance of the GP classifier model was evaluated using the receiver operating characteristic (ROC) curve [52], which plots the true positive rate against the false positive rate over a range of classification thresholds. When evaluating the GP classifier model on data with known class labels, a true positive (TP) occurs when an experiment is actually non-fouling ($c = 1$) and is correctly predicted as non-fouling ($\hat{c} = 1$). A false positive (FP) occurs when an experiment is predicted as non-fouling ($\hat{c} = 1$) but is actually fouling ($c = 0$). Similarly, a true negative (TN) occurs when an experiment is correctly predicted as fouling ($c = 0, \hat{c} = 0$). A false negative (FN) occurs when an experiment is predicted as fouling ($\hat{c} = 0$) but is actually non-fouling ($c = 1$). The true positive rate (TPR) indicates the number of actual non-fouling cases that are correctly predicted:

$$TPR \equiv \frac{TP}{TP + FN} \quad (22)$$

The false positive rate (FPR) indicates the number of fouling cases that are incorrectly predicted as non-fouling:

$$FPR \equiv \frac{FP}{FP + TN} \quad (23)$$

For an ideal classifier, TPR is 1 and FPR is 0. The area under the ROC curve (AUC) was used as the primary metric to assess the overall classification performance. A higher AUC value indicates a better ability of the model to distinguish between feasible and infeasible conditions. In addition, the difference between TPR and FPR, a metric known as Youden's J statistic or Youden index [53,54], was used to determine the optimal value of the classification threshold ϕ_c , which was then applied to assign final class labels during model deployment.

4. Results and discussion

A total of 50 experiments were carried out completely autonomously by the experimental platform to develop a classifier model in a closed-loop setup that can detect fouling-free feasible conditions in the precipitation process. These experiments were grouped into three types of experimental designs: (i) DoE (Experiments 1–8), (ii) class balancing experimental design (Experiments 9–14), and (iii) AL experimental design (Experiments 15–50). All of these were autonomously designed and executed by the experimental platform, as shown in Fig. 8. The 50-experiment campaign was completed in about five days, enabled by the high throughput of the platform and the rapid kinetics of precipitation. A sensitivity assessment of the fouling classification with respect to the selected constriction threshold (20%–70% diameter reduction) is provided in the Supplementary Information (Section S2), confirming that moderate variations in the threshold do not alter the classification of the initial DoE experiments, including both feasible and infeasible cases.

As shown in Figs. 8 and 9(a), the Sobol sampling method ensured a good spread of the experimental points throughout the entire range of the three input variables. However, the outcome of these eight experiments resulted in a slightly imbalanced dataset, with three infeasible and five feasible points (see Fig. 9(a)). The class distribution following this stage is shown in Fig. 10a. The scatter plots in this figure depict the multivariate distribution of the input variables along with the associated class labels (feasible or infeasible). The vertical histograms in each subplot represent the marginal distributions of the individual input variables, whereas the horizontal histograms show the marginal distribution of the class labels.

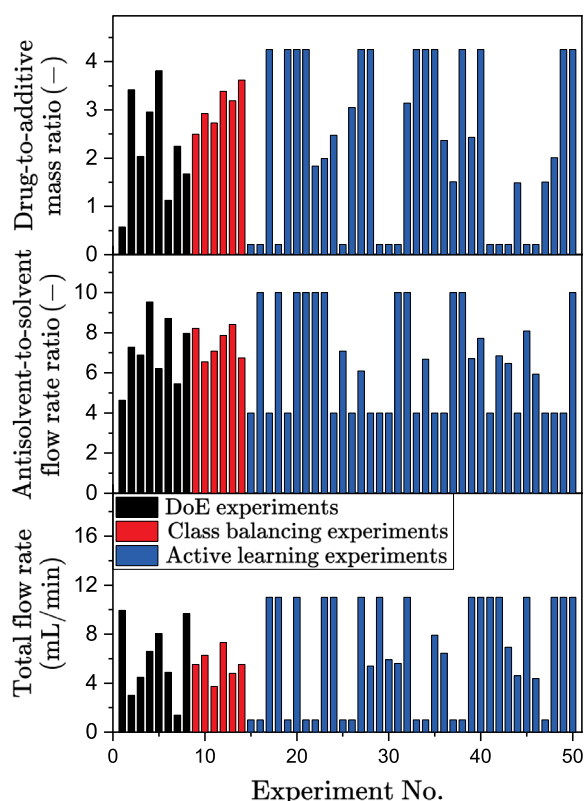


Fig. 8. The experimental campaign consisting of 8 DoE (1–8), 6 class balancing (9–14) and 36 AL (15–50) experiments used to develop the GP classifier model.

As can be seen in Fig. 10a, the proportion of the minority class (infeasible experiments) was found to be below 0.4. This proportion, defined as the ratio of the number of instances of the minority class to the total number of data points, was the threshold that we selected to indicate class imbalance. Due to this, the platform automatically triggered the next campaign, which was aimed at balancing the dataset through the class balancing experimental design algorithm described in Section 3.7.

In the first iteration of the class balancing experimental design algorithm, six new experiments were designed. These experiments are shown as red vertical bars in Fig. 8 and as “new samples” in the scatter plot of Fig. 9(e). Since there were three minority class data points (infeasible experiments) from the DoE stage (see Fig. 9(a)), the class balancing algorithm constructed a simplex geometry where at least one minority vertex was present on each of its six edges. The midpoints of all six edges were selected as the next experimental conditions. Hence, the first iteration of the class balancing experimental design involved six experiments. The various steps involved in the class balancing experimental design are illustrated in Fig. 9. These experiments were selected and executed autonomously by the platform. They resulted in five infeasible and one feasible conditions. As a result, the updated dataset contained six feasible and eight infeasible experiments, balancing the class distribution. Fig. 10b clearly shows this more even class balance through the updated histograms. At that point, since the proportion of the minority class had reached the required level (≥ 0.4), the platform automatically moved on to the next campaign, the active learning stage.

As outlined in Section 3.8, the AL approach aims to improve the classifier model by selecting the most informative experiment, that is, the condition where the model shows the highest prediction uncertainty. After the DoE stage (experiment 8), the platform calculated

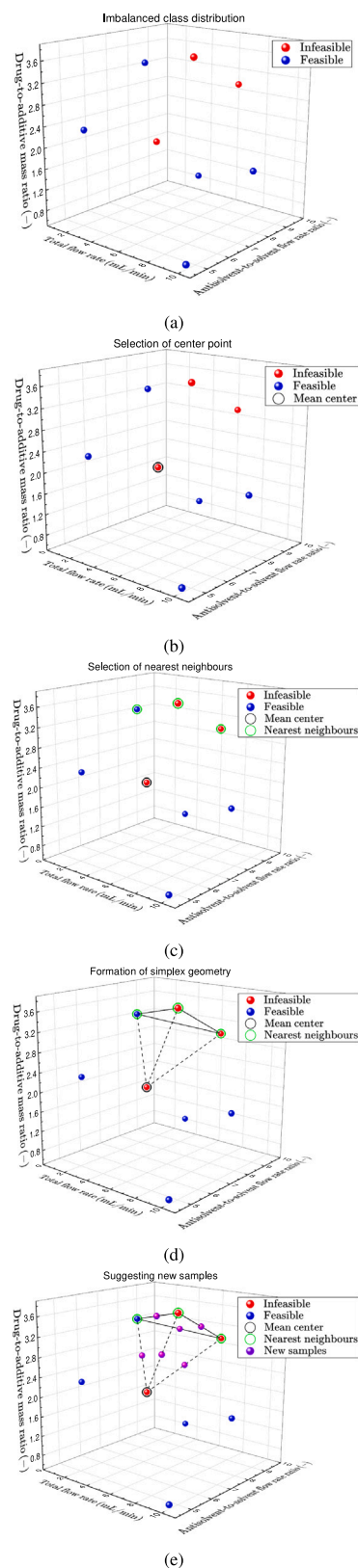


Fig. 9. A schematic illustrating the operation of the class-balancing experimental design algorithm. (a) Occurrence of imbalanced class distribution after the DoE, (b–e) Steps of the class-balancing experimental design algorithm: (b) Step 1; (c) Step 2; (d) Step 3; (e) Step 4.

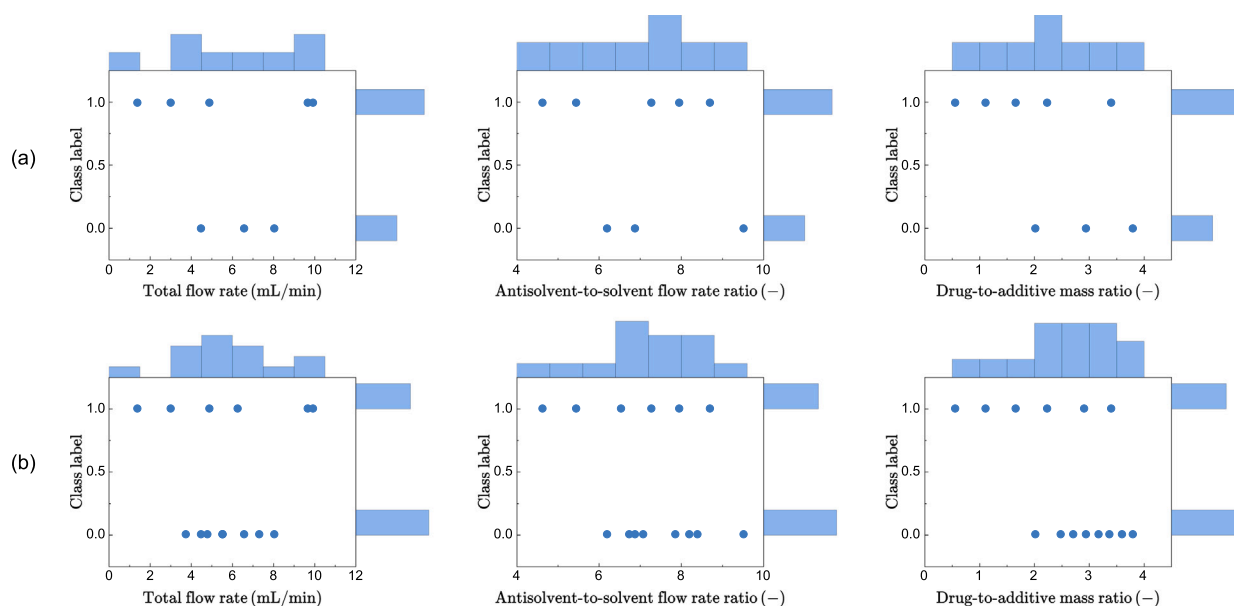


Fig. 10. Marginal histograms of inputs and class labels showing the class distribution, (a) before and (b) after the execution of class balancing experimental design.

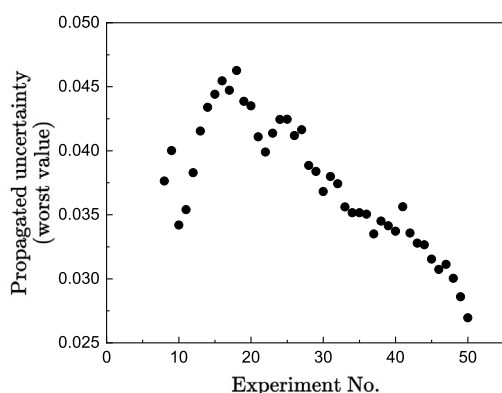


Fig. 11. The posterior propagated uncertainty (worst value across the design space) of the GP model over the course of 50 experiments.

the maximum uncertainty of the predictions of the model, defined as the largest variance in the probabilities of the predicted class over the design space. This value was approximately 0.037, as shown in Fig. 11.

During the class balancing campaign, the uncertainty initially increased, but with the progression of the AL experiments it gradually decreased. A user-defined threshold of 0.01 was set as one of the stopping criteria for the experimental process. However, when the total number of experiments reached the maximum budget of 50 (also set by the user), the platform automatically stopped the experiment execution. At that point, the maximum prediction uncertainty had dropped to around 0.025, a reduction of about 40% compared to the initial value of 0.043 before starting the AL phase. This reduction, clearly illustrated in Fig. 11, shows that the AL approach was effective in improving the confidence of the model. To further highlight the significance of these results, we provide a comparison with results obtained from simulation studies with neither class balancing nor AL experiments in the Supplementary Information (see Section S5). It is known that for a GP model the posterior uncertainty tends to increase at points that are far from the training data [37]. In this work, we employed the uncertainty propagated from the posterior latent GP model to the probability space as the objective function of AL, which is maximised. Thus, in each AL iteration, by maximising the propagated uncertainty

AL objective function of Eq. (21), we seek the new training point for which the posterior GP latent function is largely uncertain, and for which the classifier assigns equal probabilities to the classes (points with largest entropy). Under this setting, new samples are selected with respect to the current decision boundary, but not necessarily in its immediate vicinity; instead, they are chosen at locations where the latent GP function exhibits high uncertainty. Iteratively picking points based on this AL objective function results in a decrease in posterior uncertainty over the course of the experiments.

It is important to note that the AL procedure follows a sequential approach. After every new experiment, the classifier model was re-trained using all available data points to identify the next most informative experiment. Therefore, the standard model validation approaches using a train-test split was not included in the AL loop. Instead, the quality of the model in terms of classification performance (prediction accuracy) was monitored separately by the platform.

After completing all 50 experiments, the entire dataset was used to evaluate the classifier model. A cross-validation approach was followed, in which the entire dataset was randomly split into training and test sets (80% training and 20% testing) for five different random splits. The performance of the classifier was then evaluated on each test set using the area under the ROC curve, commonly known as the AUC score. An AUC score equal to 1 indicates perfect classification performance, where the model correctly predicts the class labels of the test data (see Section 3.9). As shown in Fig. 12a, the AUC scores for the five cross-validation runs were greater than 0.8, indicating strong and consistent model performance. During cross-validation, five ROC curves were obtained. In each case, the optimal classification threshold – defined as the value that maximises the difference between TPR and FPR, also known as Youden's J statistic or Youden index [53,54] – was found to lie within a narrow range of $0.49 < \text{threshold} < 0.51$. Hence, after the cross validation, an optimal threshold of 0.5, i.e., $\phi_c = 0.5$, was chosen for predicting feasibility.

An advantage of this validation approach is that it checks how well the model performs across different train-test splits. Consistently high performance across all these splits suggests that the dataset used for training is of high quality, both in terms of diversity and information content. This makes it easier for the model to learn the general behaviour or pattern of the system. This also highlights a key benefit of AL: it can help generate small but high-quality datasets that are sufficient for training data-efficient ML models, such as GP models.

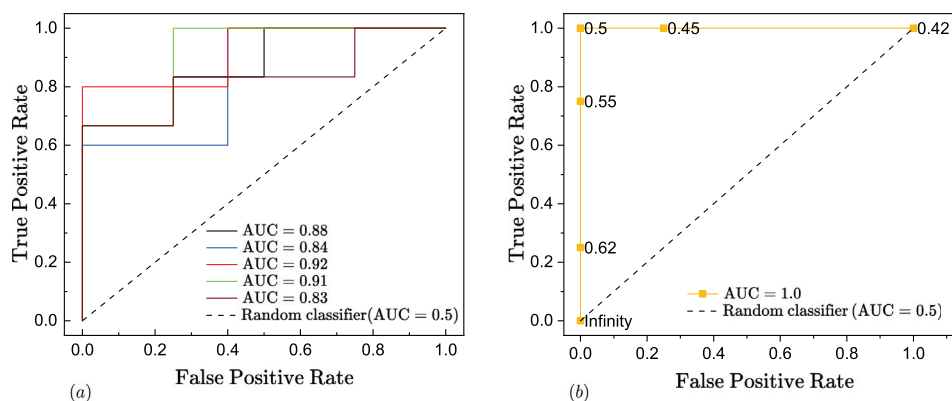


Fig. 12. ROC curves for (a) 5-fold cross-validation on the full dataset, and (b) a model trained on the full dataset and validated using eight additional experiments. The ROC curve in (b) also shows the different classification thresholds such as 0.42, 0.45, 0.5, 0.55 and 0.62. The *infinity* point in the curve refers to the point where both true positive rate and false positive rate are zero.

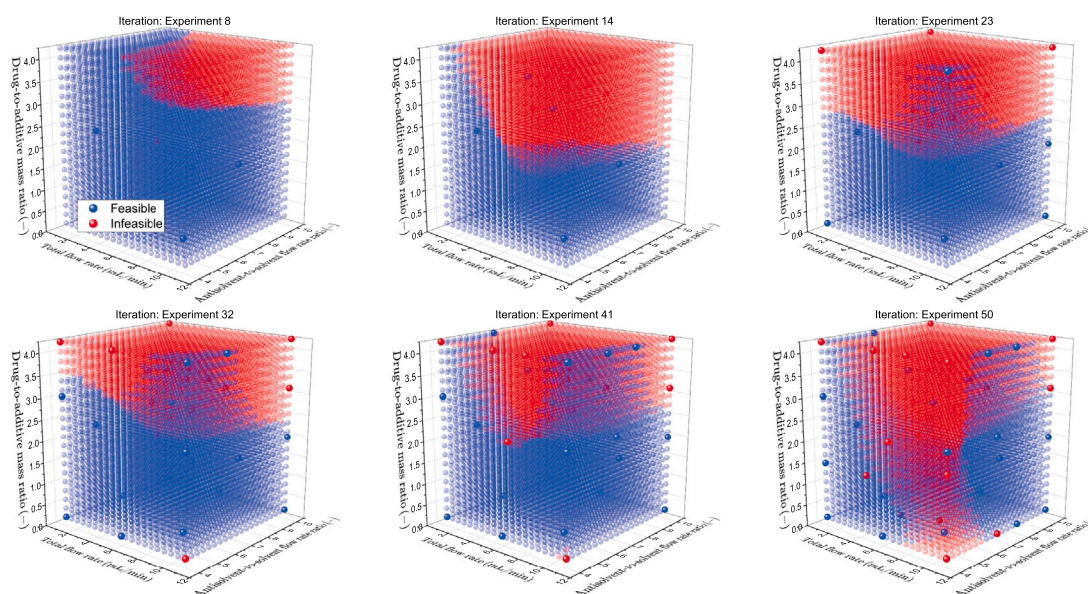


Fig. 13. Feasibility maps of the GP classifier model developed over the course of 50 experiments. The training points used at each stage are highlighted as filled spheres in the plots.

To further assess the reliability of the classifier, eight additional experimental conditions were manually selected. These were chosen by examining the feasible region predicted by the classifier model trained using the whole 50 experimental data and with a classification threshold of 0.5 (i.e., $\phi_c = 0.5$, obtained from the cross validation step). Using the predicted region, we selected the eight new experimental conditions: four expected to be feasible and four expected to be infeasible. When these experiments were executed on the actual platform, the experimental outcomes matched the expectations exactly. All four predicted feasible points were indeed feasible, and the four predicted infeasible points were confirmed as such. This result is also evident in the ROC curve shown in Fig. 12b. As stated above, this curve reflects the classifier model trained on the 50 data points and tested on the eight additional validation points. The resulting AUC score was 1, indicating perfect classification performance on the new experiments and confirming that the probability threshold of 0.5 is an optimal value (see Fig. 12b, which shows that at 0.5 threshold, the TPR is 1 and the FPR is 0). Additional classification metrics such as confusion matrices, precision scores and recall scores at different values of classification threshold can be found in the Supplementary Information (see Section S4). To assess longer-term robustness, the four operating conditions

classified as feasible were operated continuously for up to 2 h. In all cases, no sustained pressure increase was observed, confirming that, for these conditions, the fouling-free classification was maintained over the investigated duration. This result should not be interpreted as evidence of fouling-free operation over indefinitely long periods; if a more conservative feasibility map is required, longer experimental durations should be selected accordingly.

Finally, based on the 0.5 probability threshold, the evolution of the predicted feasible region throughout the experimental campaign is shown in Fig. 13. This figure includes 3D scatter plots that highlight how the feasible operating space changed as the model improved. In these plots, the experimental data used for model training at each stage of the campaign are also shown as large scatter points. Furthermore, Fig. 14 illustrates 2D slices of the final predicted feasible region, with the additional validation experiments clearly marked. Representative DLS measurements for the selected feasible operating conditions confirmed nanoscale particle formation; samples were manually withdrawn and analysed offline. Full experimental conditions, dilution procedure, and stability data are provided in the Supplementary Information (Section S3).

The feasible (non-fouling) region transforms significantly as the ASR increases, from primarily occupying low flow rate areas at ASR=4, to

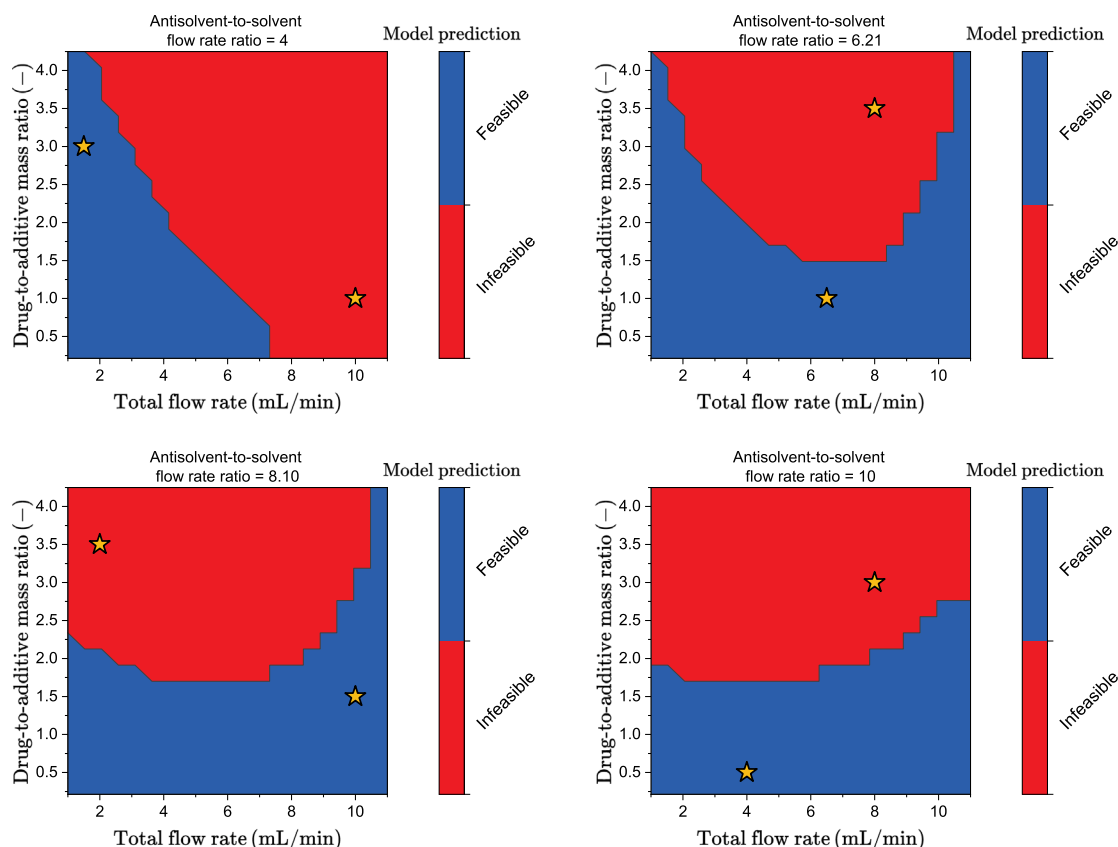


Fig. 14. Contour plots showing the feasible and infeasible regions predicted by the GP classifier model developed at the end of the experimental campaign, along with the conditions of the eight validation experiments (shown as stars in the plots).

forming characteristic U-shaped regions at intermediate ratios (5.25–9.05), and finally dominating the lower drug-to-additive ratio region at ASR=10. These intricate patterns reveal the complex interplay of multiple mechanisms in continuous antisolvent precipitation, spanning fluid dynamics (micromixing rate, shear stress distribution), thermodynamics (supersaturation level), kinetics (nucleation and growth rates), surface effects (heterogeneous nucleation), and stabilisation dynamics (additive adsorption). The relative dominance of these mechanisms shifts across the design space, creating the distinctive patterns observed in our feasibility maps.

A simple physical interpretation could suggest that high drug-to-additive ratios are infeasible because stabiliser coverage cannot keep pace with the generated surface area under strong supersaturation conditions. Amphiphilic polymers such as PVP/VA are known to inhibit nucleation and stabilise poorly soluble drugs, particularly above their critical aggregation concentrations [55–57]. However, increasing polymer concentration may also modify solution viscosity and transport characteristics, which can influence deposition behaviour in flow systems [58,59].

The U-shaped window at intermediate ASR aligns partly with mixing and shear considerations: high flow rates promote micromixing and wall shear, dispersing deposits when additive levels are sufficient. Similar reductions in deposition with increasing crossflow or shear have been widely reported in membrane and crossflow systems [60,61]. However, the feasibility observed at low flow rates at the lower end of the ASR range, even at relatively high drug to additive ratios, is not solely attributable to mixing and shear effects. At low ASR, the lower supersaturation suppresses the formation of new nuclei and slows the rate at which total particle surface area is generated, which reduces the stabiliser demand and may contribute to the observed feasibility at low flow rates. In crystallisation and precipitation systems, stronger supersaturation is consistently associated with accelerated bulk precipitation and increased fouling rates [57,62].

These interpretations are qualitative and intended to provide physical context; a quantitative mechanistic explanation would require a mathematical framework capable of resolving the fluid dynamics and micromixing, as well as all the processes relevant to particle synthesis, such as nucleation and growth, and to particle deposition and accumulation at the system walls. Moreover, such a framework would have to rely on accurate kinetic constitutive relations for each of these processes, which are typically unknown and difficult to determine. Addressing these challenges is beyond the scope of the present work and, indeed, motivated the adoption of a purely data-driven approach. The intricate behaviour observed across our design space underscores the limitations of physics-based modelling approaches for such complex systems, where multiple phenomena create emergent patterns that are challenging to predict from first principles alone. The complex interplay between operating conditions influences fouling behaviour in continuous antisolvent precipitation and validates the decision to employ a data-driven approach using GP classification and AL to map the feasible operating space without requiring detailed mechanistic understanding of each competing process, providing a practical guide for process development.

5. Conclusions

This study successfully developed and demonstrated a closed-loop autonomous screening methodology for identifying non-fouling operating conditions in the continuous antisolvent precipitation of ketoprofen nanoparticles in a CIJR. By integrating real-time pressure-based fouling detection with adaptive experimental design, combining class balancing, AL, and GP classification, the system autonomously executed 50 experiments, efficiently navigated a complex design landscape and mapped feasible regions with minimal experimental burden. This approach enabled the development of a robust continuous process without

requiring detailed mechanistic understanding of the underlying fouling phenomena.

Class balancing ensured a representative distribution of fouling and non-fouling cases within the training dataset, supporting consistent AUC scores above 0.8 across multiple validation folds. In addition, AL reduced experimental burden while maintaining high prediction accuracy, achieving an AUC of 1 on eight additional validation experiments. Reliable operating windows for continuous ketoprofen precipitation were identified, and pressure-based fouling detection was validated as an effective real-time monitoring tool for identifying fouling onset, providing practical starting points for further process development and optimisation.

Together, these outcomes establish a systematic, autonomous framework for preventing fouling in continuous pharmaceutical precipitation processes. The data-driven framework can be readily adapted to other API systems and reactor configurations by retraining the classifier on system-specific data, supporting broader implementation of autonomous continuous manufacturing strategies. Future work can build upon these findings by integrating additional process analytical technologies to monitor multiple critical quality attributes while maintaining fouling-free operation.

CRedit authorship contribution statement

Ishaa Mane: Writing – original draft, Investigation, Conceptualization. **Arun Pankajakshan:** Writing – original draft, Software, Conceptualization. **Sayan Pal:** Conceptualization. **Maximilian O. Besenhard:** Writing – review & editing, Supervision. **Asterios Gavrilidis:** Writing – review & editing, Supervision. **Federico Galvanin:** Writing – review & editing, Supervision, Conceptualization. **Luca Mazzei:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cej.2026.175671>.

Data availability

Data will be made available on request.

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